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Optimal design of European supply chains for carbon capture and storage from industrial emission sources including pipe and ship transport

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ABSTRACT

Carbon capture and storage technologies are key to remove carbon dioxide emissions from hard-to-decarbonise industries, such as steel, cement and refining sectors. A multi-echelon mixed integer linear programming model is developed for the optimal design of carbon capture and storage supply chains from industrial sources at a European level.

The model is based on exact coordinates and comprises all the stages of a carbon capture and storage chain in Europe, including multiple capture plants across industrial sources, CO_2 transport through pipelines or ships to existing docks, onshore and/or offshore geological storage basins. The optimal infrastructure is optimised in economic terms by pursuing either country-wise or Europe-wide carbon reduction targets.

Removing 50% of industrial CO_2 emissions in each country costs 60.5 €/t, which increases up to 81.4 €/t if onshore storage is forbidden. Variations of carbon reduction target are analysed. Although CO_2 transport by pipeline dominates in terms of volumes, ships can have an important role for Southern Europe countries, if CO_2 storage is restricted to offshore North Sea basins. The setting of a Europe-wide reduction target produces a slight decrease in costs in all analysed scenarios.

1. Introduction

Over the last years there has been an increasingly urgent need to push research towards the necessity of containing and tackling the global warming threat generated by the anthropogenic emissions of greenhouse gases (GHGs), particularly of carbon dioxide (CO_2) (IPCC, 2018). Energy and industry are fundamental players in this environmental challenge, considering that these two sectors are responsible for 60% of anthropogenic CO_2 emissions worldwide (Nations, 2015). However, if on the one hand the transition of power sector towards renewables or, more in general, low carbon sources is technically feasible and likely to occur in a relatively short term time horizon, oppositely the decarbonisation of industry and industrial clusters cannot rely on scenarios that do not contemplate, at some point in time, an extensive use of carbon capture and storage (CCS) technologies applied to high-purity CO_2 sources (Leeson et al., 2017). Especially in the European context, as technological upper limits are being reached in terms of efficiency, CCS is becoming widely acknowledged as the unique option to achieve a significant and long-term decarbonisation of such industrial carbon-intensive sectors (European Commission, 2020). For such reasons, this study will only focus on industrial emission sources.

The design of CCS infrastructures is characterised by an intrinsic combinatorial complexity comprising a multi-echelon array of technological options, from the different capture plants to avoid the CO_2 emissions to the variety of transport means towards sequestration into deep geological basins. Hence, a widespread and cost-effective deployment of CCS requires to be designed through quantitative modelling tools that are proper of the process systems engineering field, namely supply chain (SC) optimisation, tipically through mixed integer linear programming (MILP). MILP models are commonly regarded as powerful methodologies to optimise complex multifaceted systems, such as energy systems, and they have been recently exploited for the case of CCS networks (Tapia et al., 2018). Several studies optimised CCS SCs at region-to-country scale, e.g. Han and Lee (2012) and Lee et al. (2017) for the region of Pohang in South Korea, Elahi et al. (2014, 2017) for the United Kingdom, Kalyanarengan Ravi et al. (2016) for the Netherlands, Middleton and Yaw (2018) for Alberta (Canada). Higher-level continent-wide analyses were proposed as well. Focussing on Europe, Morbee et al. (2012) determined the transport cost of a Europe-wide CCS infrastructure with a study aimed at the optimal deployment of a pipeline-based infrastructure; d'Amore and Bezzo (2017) optimised a

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Nomenclature

Acronyms

B€	10^9 €
CCS	Carbon dioxide capture and storage
FSI	Floating storage and injection
GHG	Greenhouse gas
LNG	Liquefied natural gas
MILP	Mixed integer linear programming
SC	Supply chain

Sets

C	Country, $C = \{Austria, \ldots, United\ Kingdom\}$
$cement_n$	Cement plant, $cement_n = \{c_{1-111}\}$
k	Capture plant, $k = \{ks_{1,2,3}, kc, kr_{1,2,3}\}$
kc	Capture plant for cement plant
kr	Capture plant for refinery, $kr = \{kr_{1,2,3}\}$
ks	Capture plant for iron and steel industry, $ks = \{ks_{1,2,3}\}$
n	Node, $n = \{s_{1-25}, c_{1-111}, r_{1-59}, z_{1-53}, o_{1-34}, p_{1-31}\}$
$offshore_n$	Offshore virtual node, $offshore_n = \{o_{1-34}\}$
p	Port, $port_n = \{p_{1-31}\}$
q	Transport size, $q = \{q_{1-4}\}$
$refinery_n$	Refinery, $refinery_n = \{r_{1-59}\}$
$seque_n$	Sequestration basin, $seque_n = \{z_{1-53}\}$
$steel_n$	Iron and steel industry, $steel_n = \{s_{1-25}\}$

Scalars

α^C	country-wise carbon reduction target, [10%→73% of country emissions]
α^{EU}	European carbon reduction target, [10%→92% of Europe emissions]
$\widehat{E}^{ship}$	Emissions from ship engine, [=4.3·10^{-6} (t of $CO_2^{emitted}$)/(t of $CO_2^{transported}$)/km]
LD_{ref}^{ship}	Reference ship transport distance, [=1000 km]
$\widehat{Q}^{ship}$	Unitary ship capacity, [=10 kt of CO_2/ship]
t^{ship}	Reference roundtrip time through ship, [140 h]
USC	Unitary sequestration cost, [€/t of CO_2]
v^{ship}	Average ship speed, [=14.3 km/h]

Parameters

CCA_k	Unitary cost of CO_2 avoidance for capture plant k, [€/t of CO_2]
$CCA_{k,n}$	Unitary cost of CO_2 avoidance for capture plant k in node n, [€/t of CO_2]
$\Delta UTC_q^{ship,i}$	Percentage variations of unitary transport cost through ship for size q, [%]
fix_k^{CCA}	Fixed cost component of capture plant k, [%]
$IN_{0,k}$	Reference-case captured flowrate from capture plant k, [t of CO_2/year]
$LD_{n,n'}^{pipe}$	Matrix of distances for pipeline transport between nodes n and n', [km]
$LD_{n,n'}^{ship}$	Matrix of distances for ship transport between ports in nodes n and n', [km]
η_k	Capture efficiency of capture plant k, [%]
$\Omega_{n,n'}$	Additional cost for offshore pipeline between nodes n and n', [=1.5]
$\Psi_{n,C}$	Nodes n located in country C
Q_q^{max}	Maximum flowrate for transport size q, [t of CO_2/year]
Q_q^{min}	Minimum flowrate for transport size q, [t of CO_2/year]
ρ_k	Rate of increase in CO_2 emissions for capture plant k
UTC_q^{pipe}	Unitary transport cost through pipeline for size q, [€/km/t of CO_2]
$UTC_q^{ship,i}$	Intercept coefficient for calculation of unitary transport cost through ship for size q, [€/t of CO_2]
$UTC_q^{ship,s}$	Slope coefficient for calculation of unitary transport cost through ship for size q, [€/km/t of CO_2]
Θ_n	Additional cost for offshore storage in node n, [=2.5]
var_k^{CCA}	Variable cost component of capture plant k, [%]

Continuous variables

$IN_{k,n}$	Captured flowrate through plant k in node n, [t of CO_2/year]
IN_n^{ship}	Yearly emissions from ship transport allocated to port in node n, [t of CO_2/year]
N^{route}	Total number of routes covered by ship transport, [routes]
N^{ship}	Total number of ships employed for all routes, [ships]
$N_{n,n'}^{ship}$	Number of ships employed for the route between ports in nodes n and n', [ships]
OUT_n	Sequestered flowrate in node n, [t of CO_2/year]
$Q_{n,n'}$	Total yearly transported flowrate between nodes n and n', [t of CO_2/year]
$Q_{q,n,n'}^{pipe}$	Flowrate q transported through pipeline from node n and n', [t of CO_2/year]
$Q_{q,n,n'}^{ship}$	Flowrate q transported through ship from node n and n', [t of CO_2/year]
TC	Total cost, [€/year]
TCC	Capture cost, [€/year]
TSC	Sequestration cost, [€/year]
TTC	Transport cost, [€/year]
TTC^{pipe}	Pipeline transport cost, [€/year]
TTC^{ship}	Ship transport cost, [€/year]

Binary variables

$\lambda_{q,n,n'}^{pipe}$	1 if a pipeline of size q is installed between nodes n and n', 0 otherwise
$\lambda_{q,n,n'}^{ship}$	1 if ship transport of size q is operated between ports in nodes n and n', 0 otherwise

CCS SC taking into account power plants as unique CO_2 sources; d'Amore et al. (2021) optimised a Europe-wide CCS SC from both energy and industrial emissions, considering pipelines as possible transport means.

Ship CO_2 transport is emerging as a useful mean as it may represent in some circumstances a cost-effective alternative to offshore pipelines. In fact, the intrinsic flexibility of ship-based transport offers the opportunity to connect harbours (i.e., onshore unloading) or to link directly onshore docks to offshore sequestration basins (i.e., offshore unloading) through either injection into the well or by aim of floating storage and injection (FSI) units (IEAGHG, 2020). Several contributions analysed the costs of CO_2 transport infrastructures based on ships, which are typically constituted by a liquefaction unit, one or more buffer storage capacities (also known as intermediate storage capacities), loading and unloading facilities, and conditioning systems (IEAGHG, 2020). Some studies assessed the costs and technical aspects related to specific naval transportation routes. Aspelund et al. (2006) analysed the costs of transporting the CO_2 through ships from sources located in Antwerp

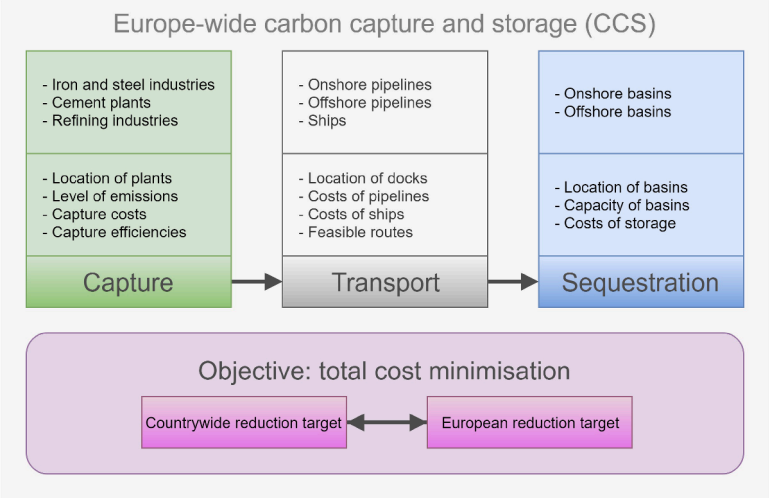


Fig. 1. Graphic representation of the CCS SC echelons considered in this study.

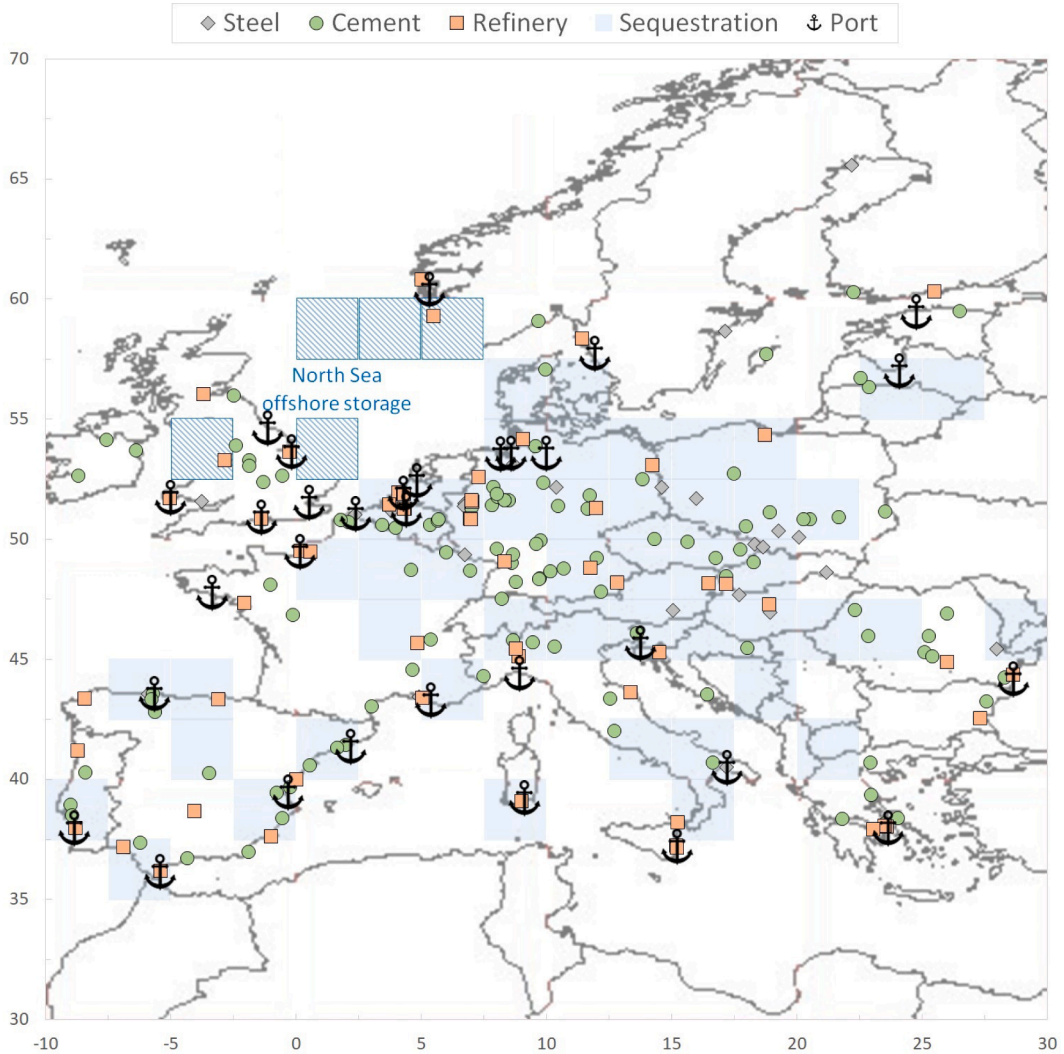


Fig. 2. Geographic location of significant industrial CO_2 sources, onshore and offshore sequestration basins and ports. Significant industrial CO_2 sources comprise steel and refining plants emitting more than 1 Mt of CO_2/year and cement plants emitting more than 0.5 Mt of CO_2/year.

Table 1
Parameter $\Psi_{n,C}$, describing which nodes n are located in country C.

Country C	Nodes n
Austria	$s10, r24$
Belgium	$s12, c2, c3, c57, c66, r11, r57, p1$
Bulgaria	$c13, r18$
Croatia	$c48, c82, r41$
Cyprus	
Czech R.	$s8, s9, c68, c73, c78, c110$
Denmark	$c1$
Estonia	$c90, p5$
Finland	$c104, r22$
France	$s22, s24, c27, c41, c81, c85, c86, c92, c93, c106, c107, r29, r31, r37, r38, r39, p11,$ $p12, p28$
Germany	$s4, s7, s13, s14, s16, s23, s25, c9, c12, c14, c19, c22, c28, c29, c37, c43, c44, c45, c49,$ $c50, c58, c59, c63, c64, c70, c72, c74, c76, c91, c95, c101, c103, c109, r2, r3, r19, r25,$ $r26, r40, r48, r49, r52, r53, r58, p2, p3, p4$
Greece	$c20, c21, c32, c42, c67, r14, r15, r21, p6$
Hungary	$s2, s21, r20$
Ireland	$c35, c51, c54$
Italy	$s19, c24, c47, c53, c94, c105, c108, c111, r1, r8, r33, r34, r35, r36, r50, p13, p14, p15,$ $p30, p31$
Latvia	$c71, p16$
Lithuania	$c52$
Luxemburg	$c84$
Netherlands	$c102, r5, r9, r54, r59, p10, p17, p18$
Norway	$c36, r7, r12, p27$
Poland	$s1, s5, s15, c7, c8, c15, c17, c18, c25, c26, r13$
Portugal	$c10, c16, c38, r28, r42, p19$
Romania	$s11, c23, c31, c39, c55, c87, c89, c100, r23, r51, p20$
Slovakia	$s18, c11, c96, r55$
Slovenia	$c79$
Spain	$s17, c5, c6, c40, c46, c56, c60, c61, c62, c65, c75, c77, c80, c97, r4, r30, r43, r44, r45,$ $r46, r47, p7, p8, p9, p29$
Sweden	$s3, s6, c4, r32, p21$
Switzerland	$c98$
U. Kingdom	$s20, c30, c33, c69, c83, c88, c99, r6, r10, r16, r17, r27, r56, p22, p23, p24, p25, p26$

(Belgium) and Brunsbüttel (Germany) to offshore unloading and injection in the North Sea; Kjärstad et al. (2016) proposed a techno-economic analysis for the Scandinavian region of ship transport for both onshore and offshore unloading; while (Bjerketvedt et al., 2020) analysed the costs of ship CO_2 transport from a cement plant located in Brevik to a harbour in Kollsnes (Norway). Differently, other contributions provided information on theoretical ship-based transport journeys, as in the reports by ZEP (2011) and by IEAGHG (2020).

However, a comprehensive design study aiming at assessing the competitiveness of ship-based CO_2 transport, and at optimising the overall multi-echelon infrastructure in a wide and integrated context is still missing. Thus, this contribution aims at providing a comprehensive SC model for the optimisation of a CCS network from European industrial emission sources (steel industries, cement plants and refineries), comprising both pipelines (onshore and offshore) and ships to existing docks to model the transport infrastructure. Additionally, alongside pursuing a Europe-wide carbon reduction target, the CCS network will be comparatively optimised for country-wise CO_2 avoidance objectives, too, in order to assess how nation-based policies may impact CCS configuration and costs with respect to a global (i.e. European) solution. The optimisation results will provide the best CCS design configurations, by determining the optimal selection, sizing and location of capture options across the different sectors, and the optimal transportation arcs and means, with particular attention to choosing pipelines or ships, to connect CO_2 sources to onshore and offshore geological sequestration basins (Fig. 1).

The next section will elucidate the modelling inputs and assumptions in terms of capture, transport and sequestration stages, following which a description of the mathematical formulation and of the investigated scenarios will be provided. Then, the results will be presented and discussed. Some final remarks will conclude the work.

2. Inputs and modelling assumptions

This study proposes a MILP modelling framework for the economic optimisation of a Europe-wide SC for CCS from significant industrial CO_2 emission sources, i.e. steel, cement and oil refining sectors (Fig. 2). All the CCS stages are included in the analysis, namely capture plants tailored for specific emissions sectors, transport through pipelines (both onshore and offshore) or ships and sequestration into appropriate geological basins (both onshore and offshore) (Fig. 1). The spatially explicit features are described through nodes $n = \{s_{1-25}, c_{1-111}, r_{1-59}, z_{1-53}, o_{1-34}, p_{1-31}\}$, comprising:

- $n = \{s_{1-25}\} \equiv steel_n$, describing 25 iron and steel industries nodes, each emitting at least 1 Mt of CO_2/year and representing 82% of European emissions from steel mills (EEA, 2020);
- $n = \{c_{1-111}\} \equiv cement_n$, describing 111 cement plants nodes, each emitting at least 0.5 Mt of CO_2/year and representing 66% of European emissions from cement sector (EEA, 2020);
- $n = \{r_{1-59}\} \equiv refinery_n$, describing 59 refineries nodes, each emitting at least 1 Mt of CO_2/year and representing 86% of European emissions from refining sector (EEA, 2020);
- $n = \{z_{1-53}\} \equiv seque_n$, describing 53 sequestration nodes, both onshore and offshore (EU GeoCapacity Project, 2009);
- $n = \{o_{1-34}\} \equiv offshore_n$, describing 34 virtual nodes characterising offshore transport routes (d'Amore et al., 2021);
- $n = \{p_{1-31}\} \equiv port_n$, describing 31 harbours for CO_2 ship transport (Eurostat, 2020; WPS, 2020).

The geographic position and characteristics of nodes $steel_n$, $cement_n$, $refinery_n$, $seque_n$ and $offshore_n$ are taken from d'Amore et al. (2021), while the location of ports $port_n$ will be discussed subsequently. This study investigates the characteristics of CCS infrastructures also at nationwide level, too; thus, European countries are listed in set C and nodes n (e.g., emission points, ports) are associated to their corresponding nation C through the parameter $\Psi_{n,C}$ (Table 1).

As for capture options, this study builds on the methodology of d'Amore et al. (2021), in which capture plants were designed through set $k = \{ks_{1,2,3}, kc, kr_{1,2,3}\}$, respectively referred to CO_2 capture from steel mills (i.e., $k = ks \equiv ks_{1,2,3}$), cement plants (i.e., $k = kc$) and refineries (i.e., $k = kr \equiv kr_{1,2,3}$). In particular, the following assessment of the different capture plants k was proposed in that analysis:

- the capture steps implemented for steel mills were: absorption capture from power plant stack (step^1), from blast furnace stoves and from coke oven flue gas (step^2), and from sinter plant (step^3) (Ho et al., 2013). These steps were combined (i.e., costs weighted on percentage contributions of the capture stages over total carbon abatement) to obtain ks_1=step^1, ks_2=step^{1+2}, ks_3=step^{1+2+3};
- the techno-economic parameters employed to describe capture technologies kc in cement plants were determined assuming to install oxy-fuel combustion capture plants (Gardarsdottir et al., 2019; Voldsund et al., 2019);

Table 2
Capture efficiencies η_k [%], rate of additional emissions due to capture operation ρ_k, costs of CO_2 avoidance CCA_k [€/t], fixed fix_k^{CCA} [%] and variable var_k^{CCA} [%] cost components, and reference-case captured flowrate $IN_{0,k}$ [Mt of CO_2/year] over technologies k. Costs in $€^{2018}$. Steel mill: ks_1=absorption from power plant, $ks_2 = ks_1$+blast furnace stoves and coke oven flue gas, $ks_3 = ks_1 + ks_2$+sinter plant. Cement plant: kc=oxy-fuel capture. Refinery: kr_1=pre-combustion capture from methane reformer, $kr_2 = kr_1$+post-combustion on power unit, $kr_3 = kr_1 + kr_2$+further emissions from other sources. Data are summarised from d'Amore et al. (2021).

Capture plant		Efficiency		Avoidance cost			Reference
	Option	η_k	ρ_k	CCA_k	fix_k^{CCA}	var_k^{CCA}	$IN_{0,k}$
Sector	k	[%]	[-]	[€/t]	[%]	[%]	[Mt/year]
Steel	ks_1	0.31	1.0	48.4	0.20	0.80	3.1
Steel	ks_2	0.62	1.2	50.6	0.22	0.78	6.3
Steel	ks_3	0.76	1.2	52.9	0.23	0.77	7.7
Cement	kc	0.90	1.0	44.4	0.83	0.17	0.7
Refinery	kr_1	0.11	1.2	42.6	0.30	0.70	0.3
Refinery	kr_2	0.33	1.2	73.5	0.57	0.43	1.0
Refinery	kr_3	0.80	1.2	89.5	0.50	0.50	2.8

Table 3
Ship transport: definition and location of ports considered in this study (Eurostat, 2020; WPS, 2020).

Node definition			Location	
Port $port_n \in n$	Name	Country	Latitude [rad]	Longitude [rad]
$p1$	Antwerpen	Belgium	0.8939	0.0765
$p2$	Bremerhaven	Germany	0.9346	0.1498
$p3$	Hamburg	Germany	0.9346	0.1744
$p4$	Wilhelmshaven	Germany	0.9344	0.1424
$p5$	Tallinn	Estonia	1.0374	0.4323
$p6$	Peiraias	Greece	0.6624	0.4125
$p7$	Algeciras	Spain	0.6306	− 0.0951
$p8$	Barcelona	Spain	0.7216	0.0378
$p9$	Valencia	Spain	0.6885	− 0.0056
$p10$	Dunkerque	Netherlands	0.8907	0.0415
$p11$	Le Havre	France	0.8634	0.0027
$p12$	Marseille	France	0.7556	0.0937
$p13$	Genova	Italy	0.7749	0.1557
$p14$	Taranto	Italy	0.7065	0.3005
$p15$	Trieste	Italy	0.7969	0.2402
$p16$	Riga	Latvia	0.9947	0.4207
$p17$	Amsterdam	Netherlands	0.9148	0.0839
$p18$	Rotterdam	Netherlands	0.9056	0.0748
$p19$	Sines	Portugal	0.6624	− 0.1548
$p20$	Constanta	Romania	0.7710	0.4998
$p21$	Gnteborg	Sweden	1.0071	0.2082
$p22$	Immingham	U. Kingdom	0.9360	− 0.0033
$p23$	London	U. Kingdom	0.8988	0.0087
$p24$	Milford Haven	U. Kingdom	0.9026	− 0.0879
$p25$	Southampton	U. Kingdom	0.8883	− 0.0244
$p26$	Tees Hartlepool	U. Kingdom	0.9530	− 0.0202
$p27$	Bergen	Norway	1.0540	0.0926
$p28$	Lorient	France	0.8334	− 0.0586
$p29$	Gijn	Spain	0.7598	− 0.0988
$p30$	Cagliari	Italy	0.6844	0.1589
$p31$	Augusta	Italy	0.6492	0.2655

Table 4
Transport parameters: ranges of admisible flowrates $[Q_q^{min} - Q_q^{max}]$ [t of CO_2/year], onshore pipeline and offshore pipeline unitary transport costs UTC_q^{pipe} [€/km/t of CO_2], and ship unitary transport cost, described through a slope $UTC_q^{ship,s}$ [€/km/t of CO_2] and an intercept $UTC_q^{ship,i}$ [€/t of CO_2] coefficients, over set $q = \{q_{1-4}\}$. Pipeline costs are taken from d'Amore et al. (2021), while ship costs are retrieved from IEAGHG (2020).

Flowrate discretisation			Pipeline transport cost		Ship transport cost	
	Range		UTC_q^{pipe}		$UTC_q^{ship,s}$	$UTC_q^{ship,i}$
Size	Q_q^{min}	Q_q^{max}	Onshore	Offshore	Offshore	
q	[kt/year]	[kt/year]	[€/km/t]	[€/km/t]	[€/km/t]	[€/t]
q_1	250	1500	0.054	0.081	0.00609	12.911
q_2	1500	2500	0.028	0.042	0.00609	11.911
q_3	2500	7500	0.016	0.024	0.00609	7.911
q_4	7500	30000	0.010	0.015	0.00609	7.911

- as for oil refineries, pre-combustion was considered to capture a quota of emissions from methane reformer (step^1) (IEAGHG, 2017), additional CO_2 was captured through post-combustion on power unit (step^2) (NETL, 2015), and further emissions from other sources (step^3) (Van Straelen et al., 2010). By combining these steps it was possible to obtain kr_1=step^1, kr_2=step^{1+2}, kr_3=step^{1+2+3}.

The detailed techno-economic characteristics of capture plants k are described in Table 2 by summarising the results reported in d'Amore et al. (2021), and following the indications given in that same article these data are employed to determine the unitary cost of CO_2 avoidance $UCC_{k,n}$ [€/t of CO_2] through capture plant k installed in node n.

2.1. Transport modes

This contribution addresses the problem of transporting the CO_2 among nodes by means of pipelines (either onshore or offshore) and ships. Offshore areas are discretised through subset $offshore_n$ (reported in d'Amore et al., 2021) and transportation arcs are considered as offshore if they fall within these areas. The possibility to store CO_2 offshore in uncertain or limited capacity basins (e.g. in the Adriatic basin Teatini et al., 2014) is not taken into account here. Differently from d'Amore et al. (2021), this study introduces the possibility of ship transport among existing European ports towards offshore storage in the North Sea area. To make the problem computationally tractable, a limited set of existing ports $n = \{p_{1-31}\} \equiv port_n$ was considered here, comprising 27 ports taken from the largest ones in terms of gross weight of goods handled (Eurostat, 2020), plus 4 additional large-scale harbours (i.e., Lorient in France, Gijn in Spain, Cagliari and Augusta in Italy) whose strategic location is dictated by the proximity of significant industrial CO_2 emission sources (WPS, 2020) (Table 3).

2.1.1. Pipeline

Pipeline option is discretised through set $q = \{q_{1-4}\}$ to account scale effects over costs due to overall transport size, and the previously mentioned set q defines the ranges $[Q_q^{min} - Q_q^{max}]$ [t of CO_2/year] of admissible flowrates. These ranges are employed to define the unitary transport costs through pipeline UTC_q^{pipe} [€/km/t of CO_2], which are calculated as in d'Amore et al. (2021), and following that previous study increased of a multiplicative factor $\Omega_{n,n'}$=1.5 in case of offshore paths between nodes n and n' (Table 4).

Table 5
Ship transport: literature review. Main inputs and outcomes from techno-economic analyses of ship-based CO_2 transport, including cases of both onshore (On.) and offshore (Off.) unloading. Buffer capacities are expressed as percentage of ship capacities. Buffer storage costs are here excluded from ship transport costs. Buffer storage costs can be accounted separately by multiplying the indicated percentage by ship transport costs. Costs are updated to €^{2018} through a sea freight transportation cost index (FRED, 2020).

				Capacities		Costs		
	Unloading		Distance	Ship	Buffer	Ship transport		Buffer
Contribution	On.	Off.	[km]	[kt]	[%]	[€/t]	[€/km/t]	[%]
Aspelund et al., 2006		v	1500	~20	150%	24.0	0.016	9.6%
ZEP, 2011*		v	180–1500	10–40	100%	14.4–17.3	0.012–0.080	5%
Kjarstad et al., 2016**	v	v	50–500	<40	–	18.3–27.5	0.069–0.550	–
Bjerketvedt et al., 2020	v		715	5	118%	33.3	0.047	9.4%
IEAGHG, 2020	v		1000	2–30	100%	24.0	0.024	8.3%
IEAGHG, 2020		v	1000	2–30	150%	30.0	0.030	6.7%

* Buffer costs included in ship transport costs and then expressed as percentage of ship CAPEX.
* Buffer costs included in ship transport costs with costs breakdown not provided.
+ The reported high cost of 0.550 €/km/t of CO_2 is for shipping 0.5 Mt of CO_2 for just 50 km.

2.1.2. Ship

Ship CO_2 transport may represent a cost-competitive alternative to pipelines and different studies assessed the techno-economic characteristics of this transport mean (Table 5). Aspelund et al. (2006) assessed the cost of a ship-based CO_2 infrastructure for offshore unloading. They analysed a 1500 km journey operated with ships of about 20 kt capacity including a buffer capacity equal to 150% of the capacity of each ship. A subsequent report by ZEP (2011) substantiated the case of offshore unloading for distances comprised between 150 km and 1500 km, by means of ships with capacity from 10 kt to 40 kt and buffer storage equal to 100% of each ship capacity. This study provided both CAPEX and OPEX, and buffer costs as a percentage of ship CAPEX. Both onshore and offshore unloading were analysed for several case studies by Kjärstad et al. (2016), though this latter study does not provide precise indications on costs breakdown. Bjerketvedt et al. (2020) focussed on onshore unloading for a 715 km path through ships with 5 kt capacity. This study incorporated operational uncertainties and fluctuations (such as sailing time and weather conditions) through a stochastic modelling framework, and showed that choosing a 118% buffer capacity would lead to a cost-effective and resilient design, although this value might increase up to 173% if acknowledging the possibility of at least one ship failure per year. A recent report by IEAGHG (2020) analysed a consistent array of case studies to determine the economic performance of a 1000 km ship-based CO_2 transport under different design configurations:

- Onshore unloading, with CO_2 transported at low pressure (7 barg, − 50 °C) and buffer storage of 100% of each ship capacity;
- Onshore unloading, with CO_2 transported at medium pressure (15 barg, − 28 °C) and buffer storage of 100% of each ship capacity;
- Offshore unloading to FSI, with CO_2 transported at low pressure (7 barg, − 50 °C) and buffer storage of 150% of each ship capacity;
- Offshore unloading to direct injection through well, with CO_2 transported at low pressure (7 barg, − 50 °C) and buffer storage of 150% of each ship capacity.

Even though highlighting that considerations on ship capacity are specific of each logistic chain, in the majority of scenarios investigated in IEAGHG (2020), the lowest cost option in terms of ship capacity was employing 10 kt-sized ships. It was shown that an increase in ship size (and unloading rate) might lead to employing less ships and decreasing their unitary cost, but the economic advantage in utilising ships larger than 10 kt of capacity could be offset by an increase in buffer storage, resulting in little change in overall cost. That study also highlighted that shipping to onshore unloading at medium pressure is more expensive than the low pressure options, as well as offshore unloading to FSI is a more costly alternative compared to CO_2 direct injection. As for intermediate storage (buffer storage is set equal to 100% of ship capacity), the model assumptions in IEAGHG (2020) are consistent with the results of Bjerketvedt et al. (2020). It turned out that shipping at least 5 Mt/year of not pre-pressurised CO_2 to onshore unloading (excluding the expenses for a subsequent pipeline towards sequestration) costs 24.0 €/t of CO_2 (comprising 2.0 €/t of CO_2 for buffer capacities), while offshore direct unloading costs 32.0 €/t of CO_2 (comprising 2.0 €/t of CO_2 for buffer capacities). These costs were found to decrease by 10 €/t of CO_2 if the CO_2 is pre-pressurised. Moreover, an analysis on the total yearly shipped capacity demonstrated, differently from pipelines, the low sensitivity of ship transport cost towards total transported quantity of CO_2 (IEAGHG, 2020).

This study incorporates the findings of IEAGHG (2020) to model ship transport cost through slope $UTC_q^{ship,s}$ [€/km/t of CO_2] and intercept $UTC_q^{ship,i}$ [€/t of CO_2] coefficients over set q, defining the same ranges $[Q_q^{\min} - Q_q^{\max}]$ of admissible flowrates already determined for pipeline transport (Table 4). It is here assumed to take into account only the shore-to-shore scenario, given its lower cost compared to offshore direct injection and the fact that the size of the problem makes it more realistic to exploit few harbours in the vicinity of offshore storage basins, which should act as collection hubs for large-scale pipelining towards sequestration in the North Sea. Unlike ship transport cost, offshore pipeline transport cost is highly dependent on the overall transported flowrate and economies of scale emerge considering that few strategic ports in the vicinity of offshore storage sites are likely to receive multiple flowrates from different projects. Moreover, ship transport costs are here modelled considering the pre-pressurised scenarios described in IEAGHG (2020), since it is here assumed to compress the CO_2 immediately after capture and costs are accordingly associated to that stage.

Overall, following the indications of IEAGHG (2020), this study assumes a buffer storage capacity equal to 100% of each ship capacity, and 10 kt of CO_2 transported per ship for each trip (i.e., the unitary ship capacity $\widehat{Q}^{ship}$=10 kt/ship). The reference ship route is designed through the following parameters to model ship transport: a roundtrip time t^{ship}=140 h, which comprises a time of 20 h for loading and 20 h for unloading, to deliver a quantity equal to 1.8 Mt of CO_2/year at a distance LD_{ref}^{ship}=1000 km (IEAGHG, 2020), hence an average (i.e., comprising also loading and unloading time) speed $v^{ship} = 2 \cdot LD_{ref}^{ship} / t^{ship}$=14.3 km/h. As for CO_2 emissions, since the tank is designed to allow an increase in pressure to balance the heat loss towards the ambient, we assume that no emissions are generated by ship transport apart from those produced by the engines. These are set equal to 1.86 t of CO_2/hour, equal to 0.65% of the transported CO_2 for a 1500 km roundtrip (Aspelund et al., 2006), which corresponds to the value $\widehat{E}^{ship}$ [=4.3·10^{-6} (t of $CO_2^{emitted}$)/(t of $CO_2^{transported}$)/km] set for calculations in this study.

3. Modelling framework

3.1. Mathematical formulation

The mathematical model is based on a MILP architecture with the objective of minimising the total cost *TC* [€/year] of the CCS system, comprising capture cost *TCC* [€/year], transport cost *TTC* [€/year] and sequestration cost *TSC* [€/year]:

$$Objective = min(TC) \tag{1}$$

$$TC = TCC + TTC + TSC \tag{2}$$

Capture cost *TCC* of Eq. (2) is evaluated as in d'Amore et al. (2021) from the captured flowrates $IN_{k,n}$ [t of CO_2/year] (i.e., carbon inlet to the transport system) through k in node n, as well as the formulation of sequestration cost *TSC* of Eq. (2) derives from the same publication and depends on the CO_2 sequestered OUT_n [t of CO_2] (i.e., carbon outlet from the transport system) in geological basin n:

$$TCC = \sum_{k,n}\left(IN_{k,n}\cdot CCA_{k,n}\right) \tag{3}$$

$$TSC = \sum_{n}\left(OUT_n\cdot USC\cdot\Theta_n\right) \tag{4}$$

where parameters $CCA_{k,n}$ [€/t of CO_2] of Eq. (3) (i.e., the node-specific cost of CO_2 avoidance through capture plant k in node n), *USC* [€/t of CO_2] (i.e., the unitary sequestration cost) and Θ_n [=2.5] (i.e., the additional cost for offshore storage in offshore basin n) of Eq. (4), are taken from d'Amore et al. (2021).

Differently from previous works, transport cost *TTC* of Eq. (2) is here determined to account the contributions of both pipelines (TTC^{pipe} [€/year]) and ships (TTC^{ship} [€/year]):

$$TTC = TTC^{pipe} + TTC^{ship} \tag{5}$$

In particular, pipeline transport cost TTC^{pipe} of Eq. (5) is calculated by taking into account scale effects over total transport capacity through discretisation set q:

$$TTC^{pipe} = \sum_{q,n,n'}\left(Q^{pipe}_{q,n,n'}\cdot LD^{pipe}_{n,n'}\cdot UTC^{pipe}_{q}\cdot\Omega_{n,n'}\right) \tag{6}$$

where $Q^{pipe}_{q,n,n'}$ [t of CO_2/year] represents the flowrate q transported through pipeline between node n and n', $LD^{pipe}_{n,n'}$ [km] is the linear distance between nodes n and n', while UTC^{pipe}_{q} represents the pipeline unitary transport cost associated to flowrate q (Table 4). The matrix of distances $LD_{n,n'}$ [km] among the nodes is calculated as in d'Amore and Bezzo (2017), i.e. by utilising the spherical law of cosines. The additional cost $\Omega_{n,n'}$ [=1.5] accounts for offshore transport operated between nodes n and n'. It is imposed a maximum transport distance through pipeline equal to $LD^{max}_{pipe} = 400$ km to reduce the connectivity among nodes and decrease the computational burden.

Ship transport cost TTC^{ship} of Eq. (5) is designed over discretisation set q and takes into account the transport distance through a linear regression of unitary costs:

$$TTC^{ship} = \sum_{q,n,n'}\left[Q^{ship}_{q,n,n'}\cdot\left(UTC^{ship,s}_{q}\cdot LD^{ship}_{n,n'} + UTC^{ship,i}_{q}\right)\right] \tag{7}$$

where $Q^{ship}_{q,n,n'}$ [t of CO_2/year] is the quantity q shipped between harbours located in nodes n and n', $LD^{ship}_{n,n'}$ is the naval distance between ports n and n' within subset $port_n$ and is retrieved from Sea-Distances (2020), whereas $UTC^{ship,s}_{q}$ and $UTC^{ship,i}_{q}$ are the slope and intercept coefficients, respectively, for the calculation of unitary transport cost through ship (Table 4). From $Q^{ship}_{q,n,n'}$, it is also possible to evaluate the number of ships $N^{ship}_{n,n'}$ that are needed for the route between harbours located in nodes n and n':

$$N^{ship}_{n,n'} = \frac{\sum_q Q^{ship}_{q,n,n'}}{\frac{8760\cdot\widehat{Q}^{ship}\cdot v^{ship}}{2\cdot LD^{ship}_{n,n'}}} \qquad \forall n,n' \in port_n \tag{8}$$

being $\widehat{Q}^{ship}$ [=10 kt/ship] and v^{ship} [=14.3 km/h] the unitary ship capacity and speed, respectively, 8760 a scalar to convert hours into years, and $LD_{n,n'}$ the sea distance between ports located in nodes n and n' within $port_n$, which must be multiplied by 2 to take into account the shipping roundtrip between travel origin and destination. Hence, the total number of ships N^{ship} employed to transport CO_2 across Europe:

$$N^{ship} = \sum_{n,n'} N^{ship}_{n,n'} \tag{9}$$

As the engines of ships generate CO_2 during navigation, it is possible to determine the emissions related to ship transport IN^{ship}_n [t of CO_2/year] from $\widehat{E}^{ship}$ [=4.3·10^{-6} (t of $CO_2^{emitted}$)/(t of $CO_2^{transported}$)/km] according to the transported quantity $Q^{ship}_{q,n,n'}$ and roundtrip distance $2\cdot LD^{ship}_{n,n'}$:

$$IN^{ship}_{n} = \widehat{E}^{ship}\cdot\sum_{q,n'}\left(Q^{ship}_{q,n,n'}\cdot 2\cdot LD^{ship}_{n,n'}\right) \qquad \forall n \tag{10}$$

Note that emissions from ship transport are allocated to the port (hence, to the country) located at the origin of the trip thus, these additional emissions are not attributed to the destination harbour, which is aimed at fulfilling storage purposes.

The flowrates discretisation is operated over $Q^{pipe}_{q,n,n'}$ of Eq. (6) and over $Q^{ship}_{q,n,n'}$ of Eq. (7) according to binary decision variables $\lambda^{pipe}_{q,n,n'}$ and $\lambda^{ship}_{q,n,n'}$, respectively, which set whether a transport mode of size q is installed between n and n' (i.e., $\lambda^{pipe}_{q,n,n'}$ or $\lambda^{ship}_{q,n,n'}$ are equal to 1 if a pipeline or ship of size q is installed between nodes n and n', 0 otherwise):

$$\lambda^{pipe}_{q,n,n'}\cdot Q^{min}_{q} \le Q^{pipe}_{q,n,n'} \le \lambda^{pipe}_{q,n,n'}\cdot Q^{max}_{q} \qquad \forall q,n,n' \tag{11}$$

$$\lambda^{ship}_{q,n,n'}\cdot Q^{min}_{q} \le Q^{ship}_{q,n,n'} \le \lambda^{ship}_{q,n,n'}\cdot Q^{max}_{q} \qquad \forall q,n,n' \tag{12}$$

where $[Q^{min}_q - Q^{max}_q]$ are the ranges of validity of possible transport capacities over set q (Table 4).

The flowrates transported through pipeline $Q^{pipe}_{q,n,n'}$ of Eq. (6) and through ship $Q^{ship}_{q,n,n'}$ of Eq. (7) allow calculating the total transported $Q_{n,n'}$ [t of CO_2/year] between nodes n and n':

$$Q_{n,n'} = \sum_q Q^{pipe}_{q,n,n'} + \sum_q Q^{ship}_{q,n,n'} \qquad \forall n,n' \tag{13}$$

which is employed to set the mass balance between nodes n and n':

$$\sum_k IN_{k,n} + \sum_{n'} Q_{n',n} = \sum_{n'} Q_{n,n'} + OUT_n \qquad \forall n \tag{14}$$

3.2. Scenarios

The model is optimised for different scenarios depending on the choice in the carbon reduction target α [%], which sets the lower bound of the CO_2 removal objective. In particular, Scenarios A optimise the network under a country-wise carbon reduction target α^C [%] equal to the 50% of the emissions from industry without capture plus those

Table 6
Scenarios A–B: description of the investigated case studies and results. Specific total cost (i.e., *TC* [€/t of CO_2]), specific capture cost (i.e., *TCC* [€/t of CO_2]), specific transport cost (i.e., *TTC* [€/t of CO_2]), specific sequestration cost (i.e., *TSC* [€/t of CO_2]), reached optimality gap (i.e., Opt. gap [%]), and solution time (i.e., Sol. time [s]). Specific costs are referred to the tons [t] of sequestered CO_2.

	Options			Results					
	Target	Sequestration		Economic results				Computational results	
	α	Onshore	Offshore	*TC*	*TCC*	*TTC*	*TSC*	Opt. gap	Sol. time
Scenario	[%]	[yes/no]	[yes/no]	[€/t]	[€/t]	[€/t]	[€/t]	[%]	[s]
A1	$\alpha^C = 50\%$	yes	yes	60.5	49.1	4.1	7.3	1.9	5738
A2	$\alpha^C = 50\%$	no	yes	81.4	49.1	14.3	18.0	2.0	9512
B1	$\alpha^{EU} = 50\%$	yes	yes	56.9	46.1	3.5	7.3	1.8	7474
B2	$\alpha^{EU} = 50\%$	no	yes	76.5	46.8	11.7	18.0	1.6	11640

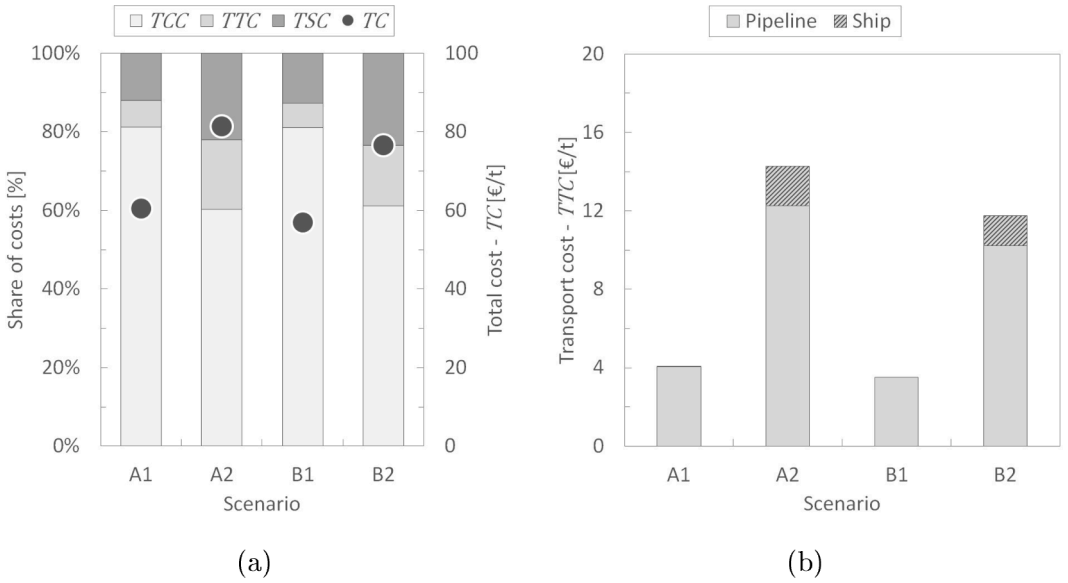


Fig. 3. Scenarios A–B: (a) share of costs (i.e., capture cost *TCC* [%], transport cost *TTC* [%], and sequestration cost *TSC* [%]) over total cost *TC* [€/t of CO_2] (the latter represented by the circles); (b) transport cost *TTC* breakdown into pipeline and ship options.

deriving from ship transport within each country *C*:

$$\sum_{k,n}\left(IN_{k,n}\cdot\Psi_{n,C}\right) \geq \alpha^{C}\cdot\sum_{n}\left[\left(IN_{n}^{max}+IN_{n}^{ship}\right)\cdot\Psi_{n,C}\right] \qquad \forall C \tag{15}$$

being $\Psi_{n,C}$ the binary parameter describing which nodes *n* are located in country *C* (Table 1).

Differently, Scenarios B optimise the CCS SC while pursuing a European carbon reduction target α^{EU} [%] equal to the 50% of the emissions from industry without capture $IN_n^{\max}$ plus those deriving from ship transport IN_n^{ship}:

$$\sum_{k,n} IN_{k,n} \geq \alpha^{EU}\cdot\sum_{n}\left(IN_{n}^{max}+IN_{n}^{ship}\right) \tag{16}$$

Moreover, additional case studies are investigated depending on the choice of including or excluding onshore CO_2 storage in the modelling framework. In fact, several European countries are currently limiting or prohibiting onshore CO_2 sequestration (Czech Republic, Germany, Poland, Sweden, the Netherlands, the United Kingdom, Austria, Croatia, Estonia, Ireland, Latvia, Finland, and Belgium) (European Commission, 2017). To take these aspects into account, on the one hand Scenario A1 and Scenario B1 are characterised by the possibility to geologically sequester the CO_2 both onshore and offshore, whereas in Scenario A2 and Scenario B2 carbon storage can only be operated offshore in the North Sea area.

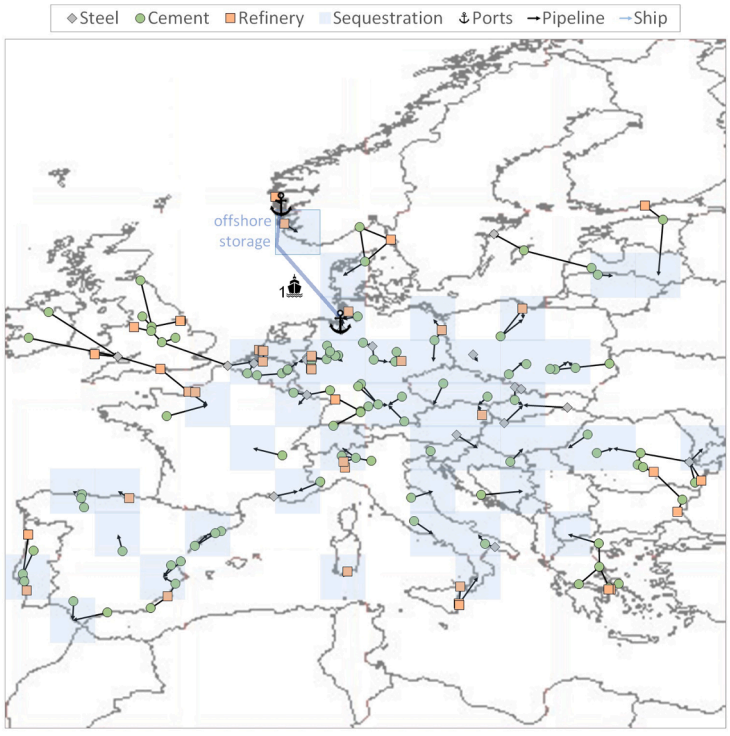
Finally, a sensitivity analysis will be performed to test the model formulation and the optimisation outcomes on variations of ship unitary transport cost and on the carbon reduction target.

4. Results and discussion

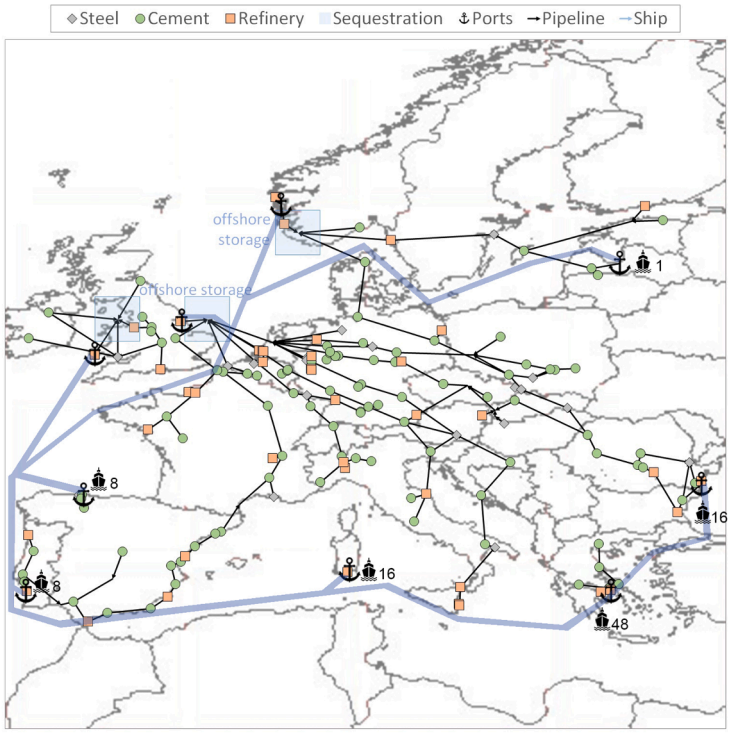
4.1. Scenarios A: Country-wise reduction target

Scenarios A aim at determining a diffused European CCS network by sequestering at least 50% of CO_2 emissions from industry in each country *C* (Table 6). Scenario A1, including the possibility of both onshore and offshore CO_2 storage, entails the lowest overall cost to install and operate the SC (60.5 €/t of CO_2), mainly constituted by capture cost (49.1 €/t of CO_2), followed by storage cost (7.3 €/t of CO_2) and transport cost (4.1 €/t of CO_2) (Fig. 3a). Both transport and storage costs are kept low by avoiding as much as possible offshore paths and basins and, as for the transport stage, ships are marginally selected in the resulting optimal infrastructure (Fig. 3b). In fact, only one ship line is exploited to transport 312 kt of CO_2/year from Bergen (Norway) to Bremerhaven in Germany (this ship accounts for less than 2% of total transported CO_2). The rest of the infrastructure is mainly based on onshore pipelines in the Continental and Mediterranean areas and few and rather short offshore pipelines where necessary, i.e. nearby the United Kingdom and in the Northern regions towards Scandinavia (Fig. 4a).

The situation drawn by Scenario A1 changes drastically when analysing the outcomes from the optimisation of Scenario A2, i.e. that one



(a)



(b)

Fig. 4. Scenarios A CCS SC configuration: (a) Scenario A1, and (b) Scenario A2.

contemplating only offshore basins as possible storage options. Despite managing to keep capture cost identical to that of Scenario A1 (49.1 €/t of CO_2), total cost is much higher (81.4 €/t of CO_2, i.e. +34.5% compared to Scenario A1), due to a noticeable increase in both transport cost (14.3 €/t of CO_2, i.e. more than tripled compared to Scenario A1) and sequestration cost (18.0 €/t of CO_2, i.e. more than doubled compared to Scenario A1) (Fig. 3a). These higher costs are due to the necessity of transporting large flowrates of CO_2 deriving from all the countries *C* towards few offshore sequestration basins located in the North Sea area, where injection is 2.5 times more expensive than in onshore sites. This complex infrastructure utilises both onshore/offshore pipelines and ships, and in fact the contribution of these latters emerges

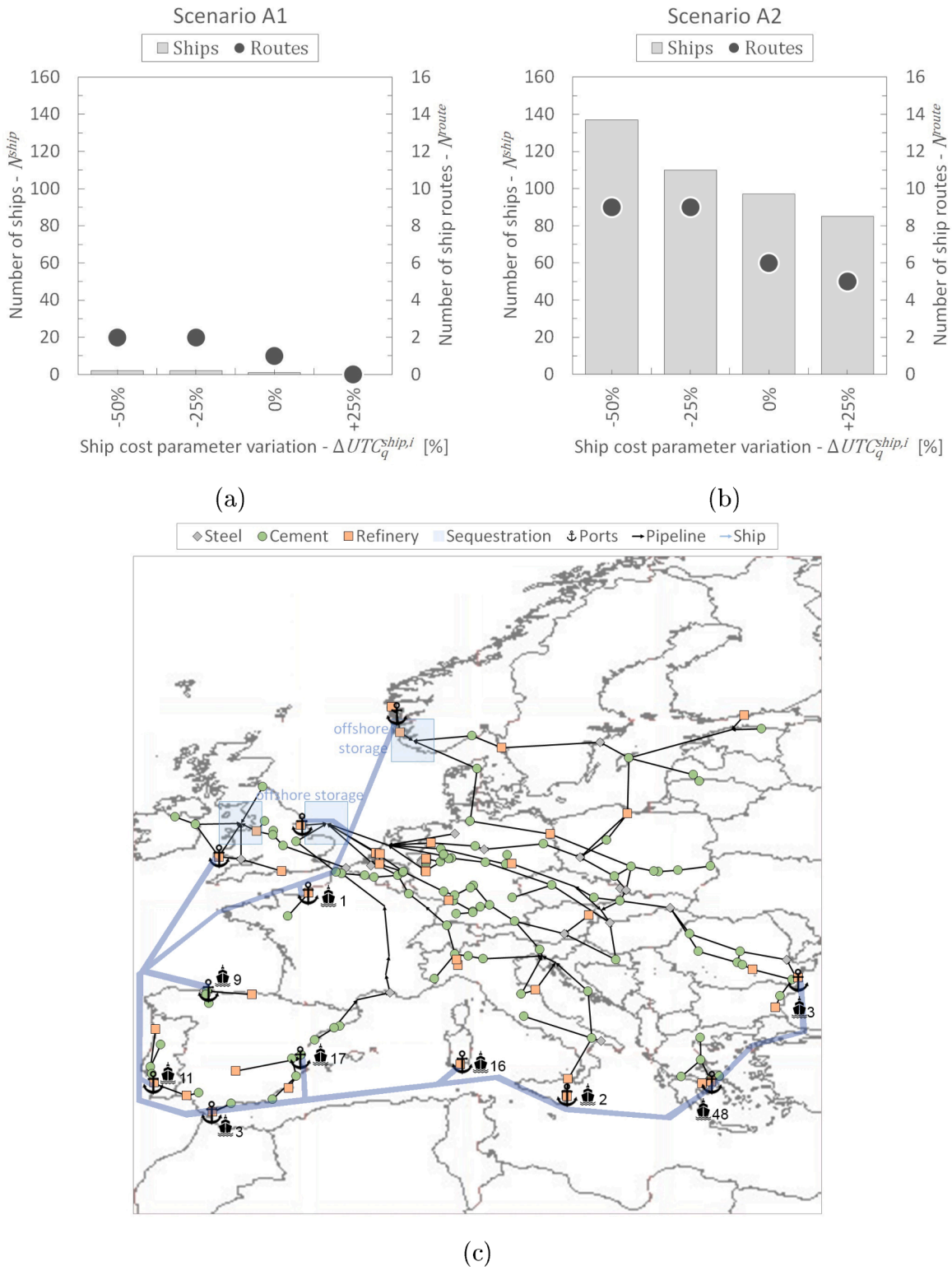


Fig. 5. Scenarios A: Results of sensitivity analyses on variations $\Delta UTC_q^{ship,i}$ [%] of ship unitary transport cost in terms of number of ships employed N^{ship} [ships], naval routes covered N^{route} [routes] and optimal CCS SC. (a) Scenario A1, (b) Scenario A2, and (c) Scenario A2 configuration for $\Delta UTC_q^{ship,i} = -25\%$.

as significant in the resulting transport cost breakdown (Fig. 3b). The optimal CCS SC suggests the exploitation of 97 ships to transport 18.6 Mt of CO_2/year (corresponding to 10.7% of total European transported flowrate) towards offshore storage in the North Sea: 48 ships from Peiraias (Greece), 16 ships from Constanta (Romania), 16 ships from Cagliari (Italy), 8 ships from Sines (Portugal), 8 ships from Gijón (Spain) and 1 ship from Riga (Latvia). The rest of the transport system is distributed across Central Europe and the Mediterranean area and converges to the available storage sites in the North Sea (Fig. 4b). The yearly CO_2 emissions generated by the engines of the ships are lower than 0.3% of the total sequestered amount. Being particularly far from the North Sea and characterised by the fact that all the CO_2 is

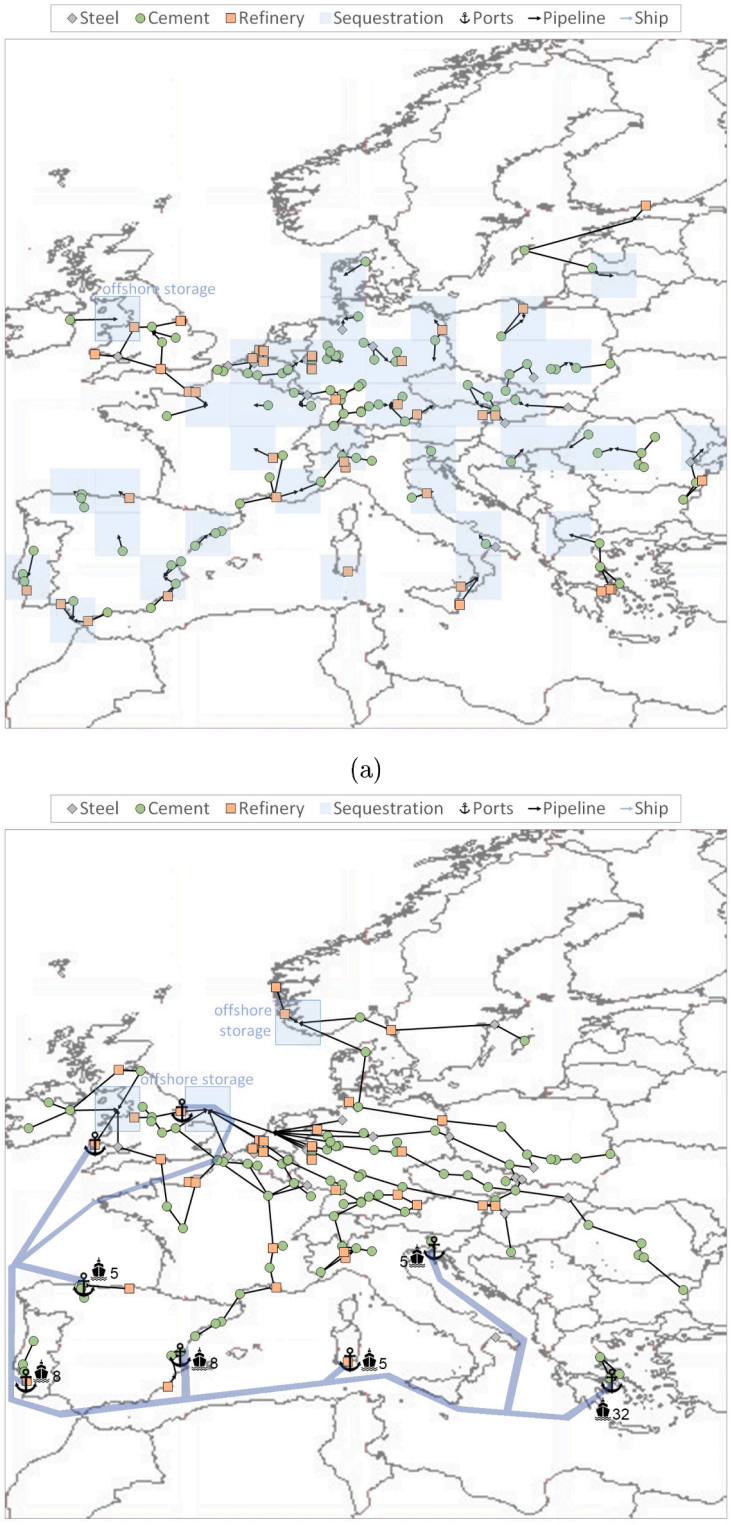


Fig. 6. Scenarios B CCS SC configuration: (a) Scenario B1, and (b) Scenario B2.

transported with ships, Greece constitutes an interesting case in point to analyse the economic results at country-wise scale between setting a onshore- or offshore-driven sequestration policy for countries in the Mediterranean area. Despite managing to keep capture costs unchanged between Scenario A1 and Scenario A2 (equal to 39.1 €/t of CO_2), the Greek transport and sequestration infrastructure is strongly penalised by the constraint of offshore storage: transport cost increases from 7.6 €/t of CO_2 (Scenario A1) to 42.5 €/t of CO_2 (Scenario A2), of which 37.1 €/t of CO_2 given by ship transport, while sequestration cost raises from 7.2 €/t of CO_2 (Scenario A1) up to 18.0 €/t of CO_2 (Scenario A2). In particolar,

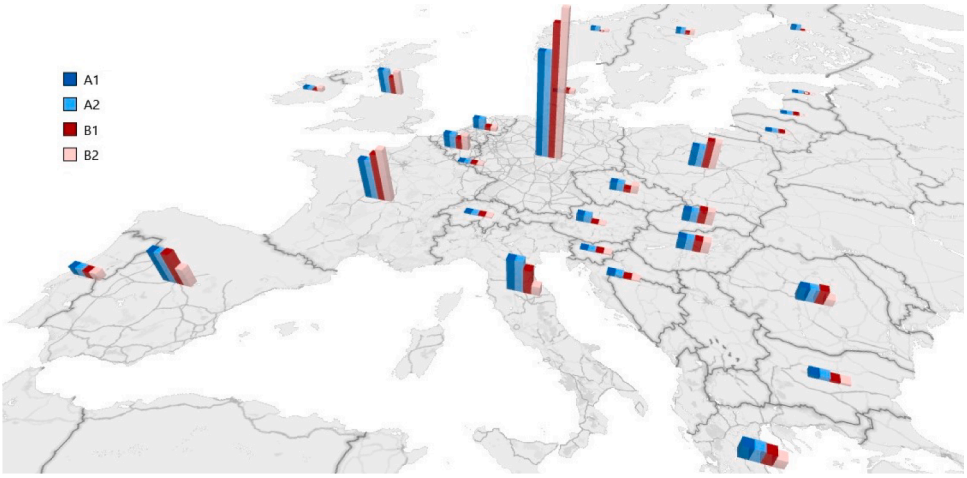


Fig. 7. Scenarios A–B: country-wise contributions in terms of yearly captured CO_2.

Greek ships transport 5.6 Mt of CO_2/year for a distance of 4797 km and emit 116 kt of CO_2/year, which corresponds to 0.07% of the European sequestered CO_2.

Being optimised for fixed country-wise carbon reduction targets, Scenario A1 and Scenario A2 entail the same results in terms of CO_2 capture. The carbon avoidance policy endorsed in these case studies determines a significant contribution from nations with a high level of emissions from industry, particularly Germany, along with others such as France, Spain, Italy, Poland and the United Kingdom.

Under our assumptions, ships cover a marginal role compared to pipelines. However, a reduction of unitary ship transport costs could be achieved if considering the use of larger vessels (though, as highlighted previously, this may leed to higher buffer costs). As a comparison, liquefied natural gas (LNG) carriers can ship up to 250000 m^3 per journey, with significant scale effects of transport size over unitary costs. For instance, it is estimated that a cost reduction of 10% can be achieved when increasing the transported LNG volumes from 140000 m^3 to 200000 m^3 (Kamalinejad et al., 2016), and similar insights emerge from studies focussed on CO_2 naval transport (Yoo et al., 2011). Here we consider the effect of the intercept of unitary cost of ship transport under variations $-50\% \leq \Delta UTC_q^{ship,i} \leq +25\%$[1]. A 25% increase in unitary ship transport cost, which could be due to the enforcement of safety measures related to naval operations, determines that either no ships are used (Scenario A1) or the number of routes and vessels is slightly reduced (Scenario A2) with respect to the previously analysed base cases. Conversely, a 25% decrease in unitary costs leads either to the establishment of a new route between Immingham in the United Kingdom and Rotterdam in the Netherlands (Scenario A1), or to a noticeable increase in both routes and ships (Scenario A2) (Fig. 5). In fact, Scenario A2 (which is based on offshore geological sequestration) for a decrease down to $\Delta UTC_q^{ship,i} = -50\%$, progressively increases the yearly number of routes covered by ships N^{route} from 6 to 9 and the number of ships employed on a yearly basis N^{ship} from 97 to 137 (Fig. 5b), with a consequent exploitation of naval lines up to 13.8% ($\Delta UTC_q^{ship,i} = -25\%$) and 15.4% ($\Delta UTC_q^{ship,i} = -50\%$) of total transported CO_2 (though with an unchanged transport cost breakdown). New ship routes link further docks located in Algeciras and Valencia (Spain) and Le Havre (France) (Fig. 5c).

[1] It should be noted that the effect of this sensitivity is quantitatively different for short and long transport distances. The reduction of 50% of the $UTC_q^{ship,i}$ parameter corresponds to a reduction of 50% of the transport cost for distances that tends to 0 km, and a reduction of 10% for distances of 5000 km (i.e., the representative distance of CO_2 transport from Mediterranean Europe to the North Sea).

4.2. Scenarios B: European reduction target

Scenarios B are designed to provide insights on optimal CCS SCs while pursuing a European carbon reduction target, rather than a country-wise objective as in Scenarios A. This choice determines a cheaper cost for the resulting networks (Table 6). In particular, Scenario B1 (which includes both onshore and offshore as storage possibilities) exhibits a total cost of 56.9 €/t of CO_2 (i.e., − 6.0% compared to the corresponding Scenario A1), which comprises an expense of 46.1 €/t of CO_2 for capture (i.e., − 6.1% compared to Scenario A1), of 3.5 €/t of CO_2 for transport (i.e., − 14.6% compared to Scenario A1) and of 7.3 €/t of CO_2 (thus, identical to sequestration cost in Scenario A1) (Fig. 3a). Differently from Scenario A1, in Scenario B1 ship transport is totally avoided (Fig. 3b) and the resulting SC makes predominantly use of onshore pipelines to transport the CO_2 for short distances towards only onshore storage basins, with the exception of minor flowrates that are set to be stored off the coasts of the United Kingdom (Fig. 6a).

Scenario B2, being constrained to make use of mere offshore sequestration in the North Sea area, clearly shows higher costs than Scenario B1, but again also for this case study the selection of a European carbon reduction target determines better economic results than its corresponding Scenario A2, in which a country-wise capture objective was set. As a matter of fact, Scenario B2 entails a total cost of 76.5 €/t of CO_2 therefore, +34.4% with respect to Scenario B1, but − 6.0% compared to Scenario A2. If on the one hand capture cost of Scenario B2 (equal to 46.8 €/t of CO_2) is not strongly affected by the policy underpinning the optimisation, on the other side both transport cost (11.7 €/t of CO_2, i.e. +234.3% compared to Scenario B1, but − 18.2% compared to Scenario A2) and sequestration cost (18.0 €/t of CO_2, i.e. +146.6% compared to Scenario B1, but equal to that of Scenario A2) change dramatically (Fig. 3a). Given the necessity of transporting the entire amount of captured CO_2 to the North Sea offshore basins, ships are employed also in Scenario B2 (similarly to Scenario A2) for covering some offshore routes (Fig. 3b): 32 ships from Peiraias (Greece), 8 ships from Valencia and 5 from Gijón (Spain), 5 from Trieste and 5 from Cagliari (Italy) and 8 from Sines (Portugal). The overall quantity transported through ships is equal to 7.9% of captured emissions thus, naval transport is exploited at a lower pace compared to Scenario A2. This result derives from the fact that the imposition of a European carbon reduction target allows for the installation of a CCS infrastructure that lays as nearby as possible to the available offshore storage sites, and accordingly the SC stages are built in the direction of the North Sea area to exploit as much as possible the cheaper cost of a relatively short transport system (Fig. 6b).

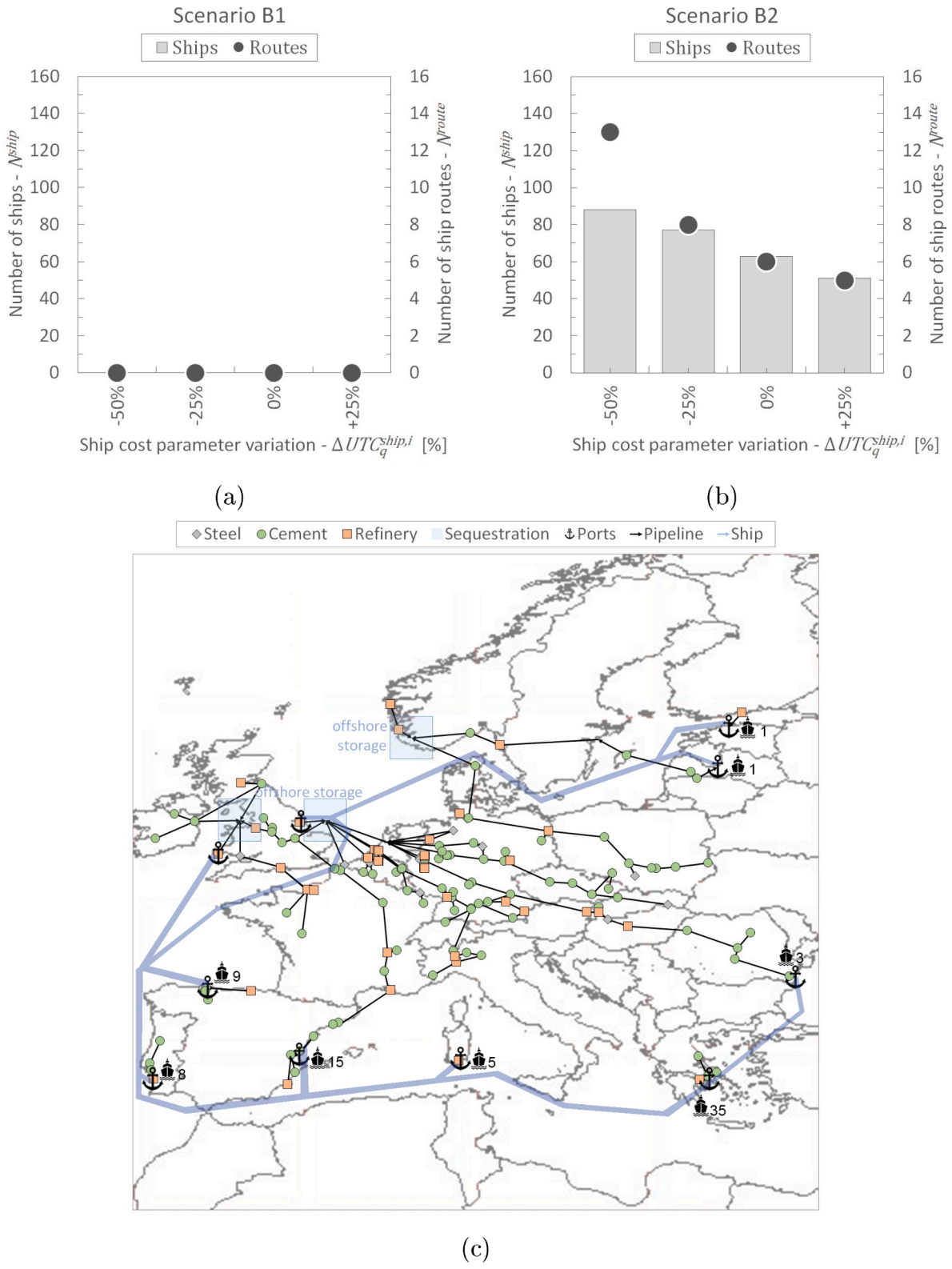


Fig. 8. Scenarios B: Results of sensitivity analyses on variations $\Delta UTC_q^{ship,i}$ [%] of ship unitary transport cost in terms of number of ships employed N^{ship} [ships], naval routes covered N^{route} [routes] and optimal CCS SC. (a) Scenario B1, (b) Scenario B2, and (c) Scenario B2 configuration for $\Delta UTC_q^{ship,i} = -25\%$.

Summarising, Scenarios B show a better economic performance with respect to Scenarios A (the decrease in unitary costs is 6.0%, with an overall saving at European level ranging between 0.9 and 1.3 B€). However, it should be highlighted that the endorsement of a Europe-wide carbon reduction target may need the establishment of complex mechanisms of transnational cooperation (d'Amore and Bezzo, 2020).

The choice of pursuing a European carbon reduction target produces a different distribution of capture plants not only with respect to Scenarios A, but also between Scenario B1 and Scenario B2 (Fig. 7). For instance, Germany, France and Poland capture more CO_2 in both

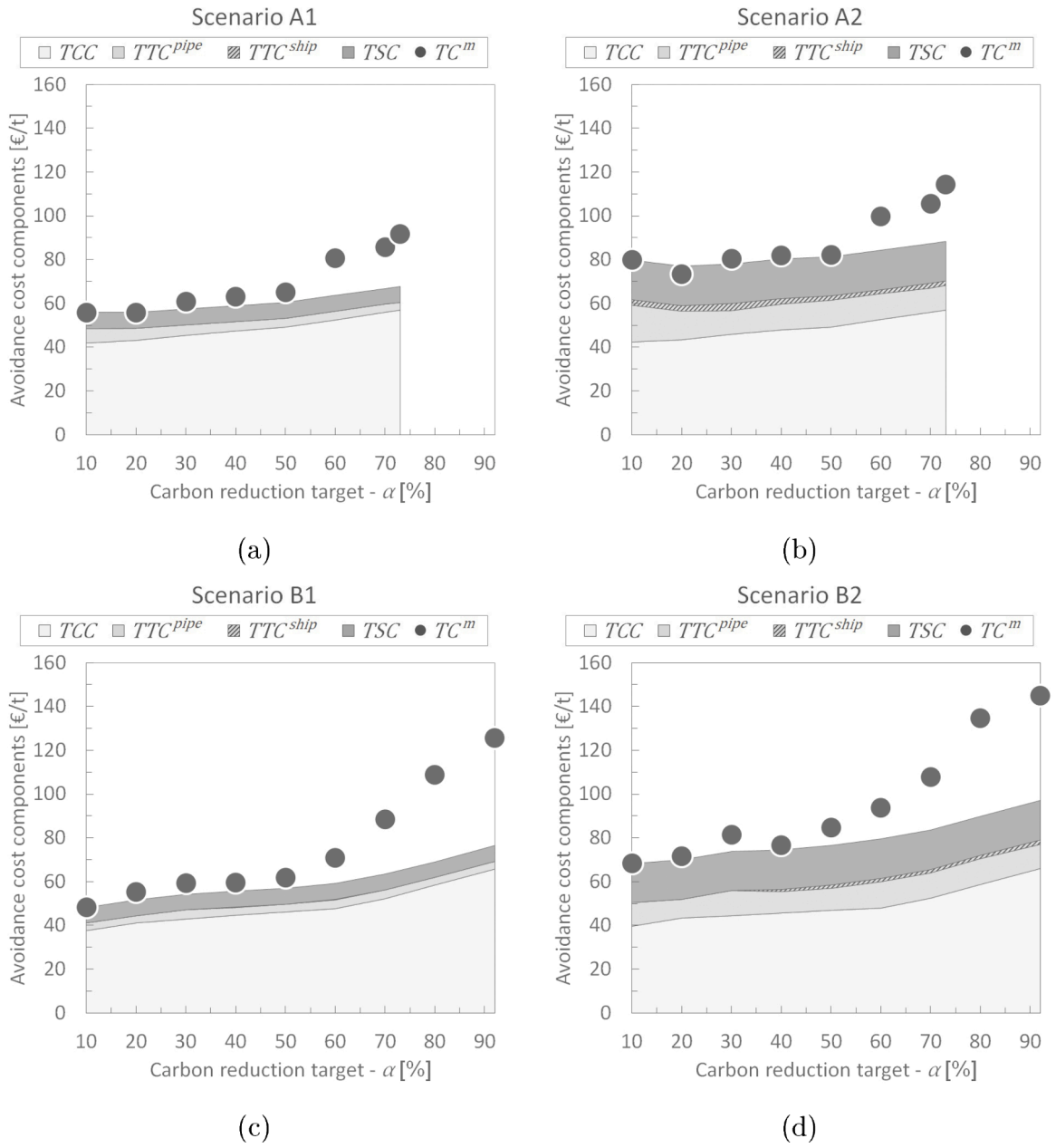


Fig. 9. Scenarios A–B: Results of sensitivity analyses on carbon reduction targets in terms of capture cost *TCC* [€/t of CO_2], transport cost through pipeline TTC^{pipe} [€/t of CO_2] or ship TTC^{ship} [€/t of CO_2], sequestration cost *TSC* [€/t of CO_2] and marginal cost of CO_2 avoidance TC^m [Δ€/Δt of CO_2] (the latter represented by the circles). (a) Scenario A1, (b) Scenario A2, (c) Scenario B1, and (d) Scenario B2.

Scenarios B than in Scenarios A, while the contribution of many Southern Europe countries, e.g. Italy, is less significant if compared with the results of a country-wise target-driven optimisation, especially in Scenario B2. In addition to this, a comparison between the results of Scenario B1 and Scenario B2 also highlights significant variations in the size of capture plants. For example, this is the case of Italy, where capture plants are installed at scale only when onshore is contemplated as storage option (Scenario B1).

Similarly to Scenarios A, a sensitivity analysis on ship unitary transport cost was performed and showed that Scenario B1 does not contemplate the use of ships as a cost effective alternative to pipelines for the entire investigated range $-50\% \leq \Delta UTC_q^{ship,i} \leq +25\%$, while Scenario B2 increases the number of routes from 6 to 13 and the number of ships from 63 to 88 (Fig. 8), determining a contribution of naval lines up to 10.5% ($\Delta UTC_q^{ship,i} = -25\%$) and 13.1% ($\Delta UTC_q^{ship,i} = -50\%$) of total transported CO_2 (with unchanged transport cost breakdown). For instance, in its $\Delta UTC_q^{ship,i} = -25\%$ case, Scenario B2 uses 15 ships from Valencia and 9 from Gijón (Spain), 5 from Cagliari (Italy), 35 from Peiraias (Greece), 8 from Sines (Portugal), 3 from Constanta (Romania), 1 from Riga (Latvia), and 1 from Tallinn (Estonia) towards Immingham and Milford Haven in the United Kingdom (Fig. 8c).

4.3. Scenarios A–B: sensitivity on carbon reduction target

The previously discussed results of Scenarios A and Scenarios B were set as reference base-cases and tested on variations of the carbon reduction targets through sensitivity analyses, to determine the marginal cost of CO_2 avoidance TC^m [Δ€/Δt of CO_2]. Preliminary analyses showed that the maximum achievable carbon reduction targets are α^C=73% for the case of Scenarios A and α^{EU}=92% in Scenarios B (Fig. 9).

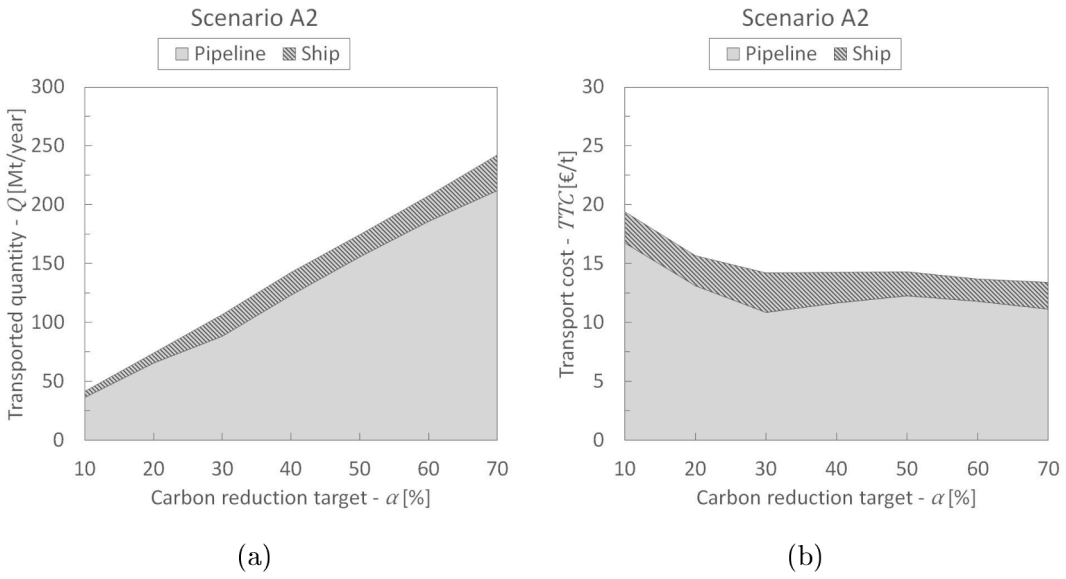


Fig. 10. Scenario A2: Results of sensitivity analysis on carbon reduction target in terms of transported quantity Q [t of CO_2/year] of CO_2 and specific transport cost TTC [€/t of CO_2] through pipeline and ship.

The upper bound α^C=73% Scenarios A was set considering that for some countries (e.g., Hungary, Czech Republic, France) CO_2 emissions are largely related to steel mills and for such plants a lower capture limit was considered with respect to emissions from other industries.

Scenario A1 and Scenario B1, i.e. those including the possibility of onshore storage, show lower total costs (with significant contributions of capture costs) for the entire range of investigated carbon reduction targets compared to Scenario A2 and Scenario B2. In particular, Scenario A1 (Fig. 9a) exhibits a total cost that varies from a minimum of 55.8 €/t of CO_2 (α^C=10%) until 67.7 €/t of CO_2 (α^C=73%), whereas Scenario B1 has slightly better economic results (Fig. 9b), with a total expenditure for establishing and operating the CCS SC increasing from 48.2 €/t of CO_2 (α^{EU}=10%), to 63.4 €/t of CO_2 (α^{EU}=70%) and until 76.4 €/t of CO_2 (α^{EU}=92%). These two case studies show also similar outcomes in terms of marginal costs of CO_2 avoidance. Scenario A1 manages to keep the marginal cost below 70 Δ€/Δt of CO_2 as long as the carbon reduction target is lower or equal than 50% of country-wise emissions (TC^m is 65.2 Δ€/Δt of CO_2 for α^C=50%), while this is true in Scenario B1 until a target of 60% of European emissions is reached (TC^m is 70.8 Δ€/Δt of CO_2 for α^{EU}=60%). Then, both case studies entail a significant increase in the marginal cost for capturing the remaining part of emissions, i.e. up to 91.6 Δ€/Δt of CO_2 in Scenario A1 and to 125.7 Δ€/Δt of CO_2 in Scenario B1 (but the latter avoids more emissions in this configuration).

When offshore is set as the unique storage option, the contribution of transport and sequestration over total cost is not negligible. Total cost in Scenario A2 (Fig. 9b) increases from 79.8 €/t of CO_2 (α^C=10%) to 88.3 €/t of CO_2 (α^C=73%), while for comparison that of Scenario B2 varies from 68.3 €/t of CO_2 (α^{EU}=10%) to 97.1 €/t of CO_2 (α^{EU}=92%), with Scenario B2 being cheaper for most part of the investigated range of carbon reduction target. Scenario A2 is characterised by a moderate drop in transport cost, which is justified by the emerging of scale effects in pipelines as larger CO_2 flowrates are transported towards the North Sea (Fig. 10). Conversely, all cost components increase almost linearly in Scenario B2 until α^{EU}=60%, and then a further growth is determined by the increasing use of ships and by capture from expensive sources to address this challenging carbon reduction target (Fig. 9d). Scenario A2 and Scenario B2 show similar marginal cost curves. In Scenario A2, TC^m increases from 79.8 Δ€/Δt of CO_2 (α^C=10%) to a maximum of 114.3 Δ€/Δt of CO_2 (α^C=73%). Interestingly, Scenario A2 shows a minimum value of marginal cost (73.4 €/t of CO_2) for a moderate reduction α^C=20%, which is due to the combined effect of the installation of the cheapest capture plants with the arising of scale effects in pipeline transport costs. On the other hand, marginal cost in Scenario B2 varies from 68.3 Δ€/Δt of CO_2 (α^{EU}=10%), to 107.8 Δ€/Δt of CO_2 (α^{EU}=70%) and up to a very high 145.0 Δ€/Δt of CO_2 to avoid the last 22% of emissions (α^{EU}=92%).

5. Conclusions

A mixed integer linear programming model was implemented and exploited to design an optimal European infrastructure for carbon capture and storage from industrial CO_2 sources (steel industries, cement plants and refineries) including both ship and pipeline transport.

The resulting minimum cost for installing and operating the network is equal to 60.5 €/t of CO_2 in the case of pursuing a country-wise reduction target of 50%; the cost is in large part constituted by capture costs, while transport and sequestration costs sum up to 18.8% altogether. These expenditures would be higher if considering only offshore storage in the North Sea, since the total cost would increase up to 81.4 €/t of CO_2 with significant shares given by the transport and sequestration stages. A cost reduction (about 6.0%) could be achieved by considering a European reduction target rather than a country-wise one, with an infrastructure which is mainly built in Northern countries in the case of offshore storage only, and reaches the Mediterranean area at significant scale just for a very high CO_2 removal objective. By varying the carbon reduction target, we were able to assess the cost of CO_2 avoidance, which can be used to obtain some insights into a possible carbon tax offsetting the costs of a European carbon capture and storage system. A carbon tax should be lower than 65 €/t of CO_2 or 85 €/t of CO_2 for capturing up to 50% of emissions depending on including or excluding onshore sequestration, respectively. Avoiding the emission to the atmosphere of the remaining part of CO_2 would lead to a further increase in the carbon tax well above 100 €/t of CO_2.

It was shown that the use of ships for transporting the CO_2, though generally of minor importance compared to the resulting vast network of pipelines, could be essential locally, especially when linking the Mediterranean area with far-distant storage locations such those in the North Sea area. In this perspective, future work will aim at assessing the exploitation of lower capacity basins in the Mediterranean (e.g., in the Adriatic Sea), which may allow reducing costs in Southern Europe, at least if a short-to-medium term horizon is considered. Finally, more detailed studies, possibly at a Country or regional level, should be car-

ried out to assess the possibility to cluster small scale emission points and to integrated existing facilities with renewable energy production (e.g., bioenergy with carbon capture and storage).

CRediT authorship contribution statement

Federico d'Amore: Conceptualization, Methodology, Investigation, Software, Writing - review & editing. **Matteo Carmelo Romano:** Conceptualization, Methodology, Writing - review & editing. **Fabrizio Bezzo:** Supervision, Conceptualization, Methodology, Writing - review & editing.

Declaration of Competing Interest

The authors declare no competing interests.

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